

MANHATTAN, KS, USA

WMO: 724555

Lat: 39.135N

Lon: 96.679W

Elev: 1056

StdP: 14.14

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 03936

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.5	(c) 8.0	(d) -5.9	(e) 4.2	(f) 5.6	(g) -1.4	(h) 5.3	(i) 9.9	(j) 24.0	(k) 48.8	(l) 20.8	(m) 44.4	(n) 3.9	(o) 10	(p) 0.522

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 22.4	(c) 99.9	(d) 75.5	(e) 96.8	(f) 75.5	(g) 93.1	(h) 74.8	(i) 78.8	(j) 92.6	(k) 77.7	(l) 91.3	(m) 76.6	(n) 89.5	(o) 11.6	(p) 200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.1	137.1	85.7	73.5	129.4	83.9	72.6	125.7	83.2	43.1	92.8	41.8	90.8	40.6	89.5	83.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
23.9	20.3	18.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
			-6.4	105.3	5.7	3.2	-10.6	107.5	-13.9	109.4	-17.1	111.1	-21.3	113.4		
			-7.0	81.8	5.6	1.3	-11.0	82.7	-14.3	83.5	-17.4	84.2	-21.4	85.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.3	30.0	33.7	44.4	54.9	65.2	75.2	80.1	77.5	69.4	56.4	43.6	32.3
	DBStd	19.88	11.02	11.63	11.38	9.63	8.56	6.89	6.29	6.35	8.55	9.61	9.88	10.50
	HDD50	2231	622	469	243	57	3	0	0	0	45	240	552	
	HDD65	5053	1086	877	643	325	109	7	1	51	297	644	1012	
	CDD50	4175	1	11	69	204	473	756	933	852	581	244	47	4
	CDD65	1523	0	0	4	22	114	313	468	388	182	31	1	0
	CDH74	17521	0	2	62	306	1226	3559	5654	4319	1999	375	19	0
	CDH80	8311	0	0	9	76	442	1630	3026	2173	849	105	1	0
Wind	WSAvg	7.4	6.8	7.5	8.8	9.5	8.1	8.0	6.7	5.9	6.5	7.0	7.1	6.7
Precipitation	PrecAvg	33.4	0.7	1.1	2.0	3.1	4.7	5.0	4.6	4.0	3.0	2.4	1.6	1.2
	PrecMax	51.9	1.6	3.3	4.7	9.3	13.1	10.2	15.5	8.5	5.3	5.6	5.0	3.7
	PrecMin	24.4	0.1	0.1	0.0	0.9	1.1	0.4	1.9	0.7	0.4	0.2	0.0	0.0
	PrecStd	6.3	0.4	0.8	1.2	1.7	3.1	2.3	3.0	1.7	1.3	1.3	1.4	1.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.4	71.6	80.5	87.6	93.1	99.8	104.4	104.3	98.9	88.6	76.9	65.8
		MCWB	49.5	54.5	60.9	66.6	71.4	75.4	75.4	74.3	72.6	69.9	59.3	57.6
	2%	DB	58.4	63.9	74.6	81.6	88.6	95.5	100.4	99.3	92.6	82.3	71.5	59.2
		MCWB	46.6	51.1	58.4	63.1	70.4	75.3	76.0	75.1	73.3	64.8	57.8	48.4
	5%	DB	52.9	57.5	69.9	77.3	84.9	92.4	97.4	94.9	89.1	78.1	65.8	54.3
		MCWB	43.5	48.7	56.0	61.3	68.6	74.8	76.3	75.1	72.2	63.4	54.7	46.2
	10%	DB	47.5	52.8	64.2	72.8	81.2	89.3	93.4	90.9	84.5	73.3	61.3	49.1
		MCWB	40.3	44.2	53.2	59.4	67.7	73.9	75.7	74.5	69.9	61.7	52.1	42.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.0	58.1	65.3	68.4	75.4	79.7	80.5	80.1	77.2	72.7	62.9	58.7
		MCDB	58.6	65.7	76.4	80.4	87.3	93.3	94.3	92.1	90.2	82.2	71.0	63.9
	2%	WB	47.4	53.4	61.8	66.1	73.1	77.8	79.1	78.2	75.4	69.5	59.6	51.4
		MCDB	56.4	61.0	69.3	76.8	84.5	91.2	93.6	91.1	88.0	78.6	66.6	55.6
	5%	WB	44.1	48.7	58.5	63.6	71.1	76.4	78.0	77.0	73.8	66.3	56.6	46.3
		MCDB	52.4	58.5	67.4	74.7	81.8	89.0	92.2	89.8	85.8	74.4	65.1	53.1
	10%	WB	40.4	44.3	53.9	60.7	69.1	74.9	76.9	75.5	71.9	63.1	52.7	42.5
		MCDB	47.9	52.1	64.4	71.2	78.7	86.4	90.4	87.8	82.4	70.8	60.4	48.9
Mean Daily Temperature Range	5% DB	MDBR	22.6	22.2	24.4	24.2	22.9	22.3	22.4	22.9	24.3	24.9	23.9	21.7
		MCDBR	32.5	32.4	33.5	30.3	27.4	25.1	26.7	28.0	26.4	30.0	31.8	30.2
	5% WB	MCWBR	21.7	21.1	19.0	16.2	12.7	10.1	8.4	9.2	10.7	15.5	19.1	20.7
		MCDWR	30.6	29.9	25.9	25.9	23.5	22.8	23.1	23.5	23.2	23.6	26.3	27.2
Clear-Sky Solar Irradiance	taub	taub	0.287	0.302	0.333	0.379	0.410	0.421	0.428	0.427	0.383	0.342	0.301	0.282
		taud	2.483	2.449	2.409	2.304	2.288	2.291	2.296	2.298	2.374	2.464	2.513	2.501
	Ebn at Noon	Ebn at Noon	284	293	293	284	277	272	269	267	273	275	276	277
		Edh at Noon	25	29	34	40	42	42	41	40	35	29	24	23
All-Sky Solar Radiation	RadAvg	736	981	1286	1631	1863	2112	2082	1858	1535	1121	808	631	
	RadStd	62	99	96	101	148	125	142	93	98	117	67	60	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (27 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page